

QUANTIZATION FOR REASONING PRESERVATION IN SMALL LLMs

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ABSTRACT

Large Language Models (LLMs) have demonstrated exceptional capabilities in textual inference, but their deployment on edge devices is limited by high computation and memory overhead. To address this, we propose LLM-QRP (**Q**uantization for **R**easoning **P**reservation), a layer-aware mixed-precision quantization framework designed to reduce model size while preserving reasoning performance. Unlike uniform post-training quantization methods, LLM-QRP identifies and preserves critical “thinking” layers by analyzing layer-wise logit evolution and entropy dynamics across transformer blocks. These signals are combined into a unified reasoning importance score that ranks layers according to their contribution during inference. We then apply an empirical mixed-precision strategy that assigns higher precision to high-scoring layers and lower precision to less important layers, targeting the Pareto frontier between model compression and reasoning performance. Experiments on multiple small models using GSM8K and TruthfulQA show that LLM-QRP reduces VRAM usage while maintaining near-baseline reasoning performance and outperforming standard uniform 4-bit quantization. The resulting models aim to provide the most compressed model at a given reasoning-performance level, enabling efficient deployment on resource-constrained hardware without significant degradation in reasoning ability.

1 INTRODUCTION

Large Language Models (LLMs) have qualitatively transformed various domains of artificial intelligence. A core advantage of local LLMs is the benefits of on-device use. It serves as an alternative pathway for LLMs to operate effectively offline, without intermediate delays from API or cloud calls, and remains beneficial for real-time, offline applications involving autonomous vehicles, financial processing, and applications requiring a fine-tuned model. However, with great model size comes a great compute cost. For instance, Qwen3.5, containing 0.8B parameters at FP16, has a memory footprint of 1.8 GB, which can adequately run inference on edge devices (Oll; Hou et al., 2025).

In post-training-quantization (PTQ), multi-step reasoning tasks are compromised first, leading to degradation in the model’s ability to reason and think (Li et al., 2026). Quantization-aware training suffers from high infrastructure costs, making compute and model training quite expensive. Recent advancements in quantization provided by GPTQ and LLM-AWQ do provide mixed-precision quantization; LLM-AWQ optimizes for which weights to preserve instead of identifying layers that actually perform reasoning (Lin et al., 2024). To address these challenges, we propose LLM-QRP (Quantization for Reasoning Preservation), a layer-aware quantization framework that explicitly targets the preservation of reasoning capability in small LLMs. LLM-QRP identifies and prioritizes “thinking layers” based on their contribution to the model’s final predictions during inference. Rather than treating all transformer layers uniformly, our framework analyzes how information evolves throughout the network and uses these signals to determine which layers are more important for preserving reasoning.

LLM-QRP combines two complementary signals for identifying reasoning-critical layers. First, we use Self-Logits Evolution Decoding (SLED), which measures how intermediate token representations evolve toward the model’s final output distribution (Zhang et al., 2025). Second, we incorporate layer-wise entropy analysis to characterize changes in the model’s uncertainty throughout the residual

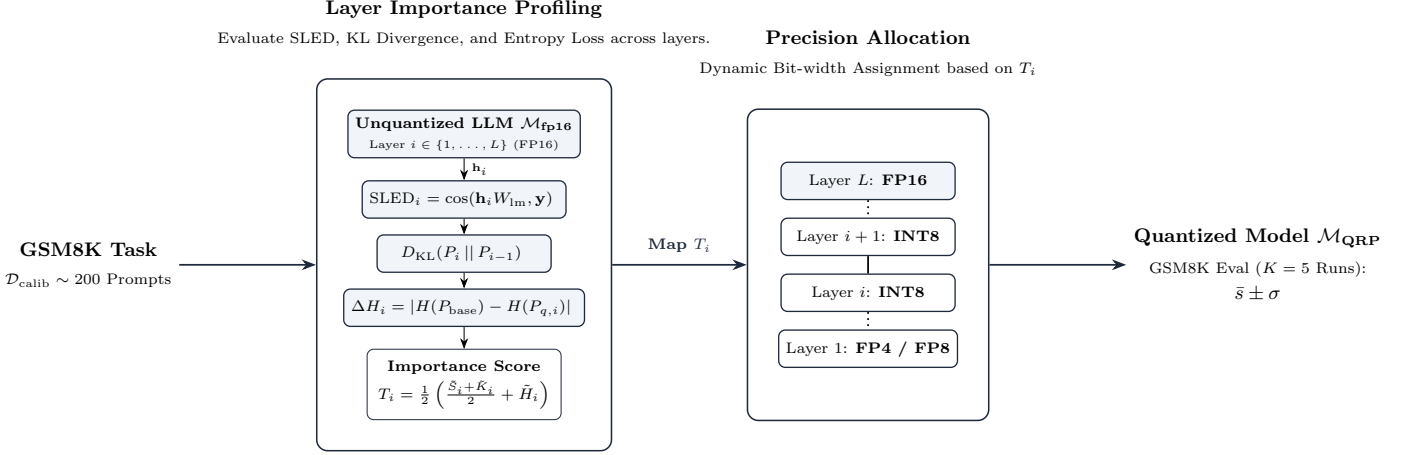


Figure 1: Architectural pipeline of the Layer-Aware Quantization & Reasoning Preservation (LLM-QRP) framework.

stream. These signals are combined into a unified layer-importance score, allowing layers to be ranked according to their estimated contribution to reasoning preservation.

The resulting layer-importance scores are then used to guide a mixed-precision quantization strategy. Rather than applying a uniform precision across the entire model, LLM-QRP allocates lower precision to less important layers while preserving higher precision for layers identified as more critical to reasoning. The resulting precision configurations are selected empirically subject to an accuracy constraint, allowing the framework to balance memory reduction against reasoning performance.

We evaluate LLM-QRP across four sub-500M parameter models and compare full-precision, standard quantized, and LLM-QRP-guided quantized variants. We further compare against existing quantization approaches, including AWQ. Our experiments demonstrate that reasoning-aware layer allocation can substantially reduce memory requirements while maintaining performance on reasoning-intensive benchmarks such as GSM8K.

Our main contributions are: (1) Proposing LLM-QRP, a layer-wise quantization framework designed to preserve reasoning capabilities in small LLMs while reducing memory requirements. (2) Combining SLED and layer-wise entropy signals into a unified layer-importance score for identifying reasoning-critical “thinking layers.” (3) Introducing a reasoning-guided mixed-precision allocation strategy that assigns precision according to layer importance rather than applying uniform quantization across the model. (4) Evaluating LLM-QRP across multiple sub-500M parameter models and demonstrating its effectiveness in balancing model compression and reasoning performance.

2 BACKGROUND

Recent work on LLM quantization has explored a range of approaches for reducing memory and computational requirements while minimizing the degradation introduced by reduced numerical precision. These methods include weight-only quantization, activation-aware quantization, mixed-precision quantization, and methods that explicitly identify sensitive weights or layers. Our approach builds upon these developments while focusing specifically on the preservation of reasoning through layer-wise importance estimation.

2.1 ACTIVATION-AWARE AND SALIENCY-BASED QUANTIZATION

LLM-AWQ is a quantization framework focusing on preserving a model’s salient weights. LLM-AWQ uses Post-Training Quantization (PTQ) to compress large language models efficiently without the immense cost of retraining. Rather than treating all weights equally, AWQ identifies activation-

aware salient weights and protects them during quantization (Lin et al., 2024). We follow the same general principle of selectively preserving important components; however, we primarily concentrate on identifying important *layers* rather than individual salient weights. Furthermore, our framework evaluates the contribution of each layer to reasoning preservation rather than relying solely on weight-level sensitivity.

SpinQuant similarly addresses the sensitivity of LLMs to quantization through rotational transformations that make the weight and activation distributions more quantization-friendly (Lee et al., 2025). By applying learned or optimized rotations, SpinQuant reduces quantization error while maintaining the underlying model capabilities. While this approach improves the numerical properties of the tensors being quantized, LLM-QRP instead focuses on determining *where* precision should be preserved in the transformer based on layer-level reasoning importance.

2.2 MIXED-PRECISION QUANTIZATION

Several works have explored mixed-precision strategies to allocate different bit-widths to different components of an LLM. SpQR introduces a sparse-quantized representation that combines low-bit quantization with a small set of higher-precision outlier weights, allowing substantial memory compression while preserving model quality (Dettmers et al., 2023b). This demonstrates that selectively allocating additional precision to sensitive components can provide a favorable accuracy–memory trade-off. However, SpQR primarily operates through weight-level outlier identification, whereas LLM-QRP determines precision allocation using layer-level reasoning importance.

AMQ explores adaptive mixed-precision quantization by allocating different precision levels according to the sensitivity of model components (Lee et al., 2025). Such approaches demonstrate that a uniform quantization configuration is not necessarily optimal across an entire model. LLM-QRP extends this principle by using SLED and layer-wise entropy sensitivity to explicitly estimate the reasoning importance of individual transformer layers before assigning their precision. Unlike AMQ, which primarily optimizes the allocation of precision according to quantization sensitivity and efficiency objectives, LLM-QRP explicitly incorporates reasoning behavior into the precision-allocation process, allowing the resulting mixed-precision configuration to prioritize layers that contribute most strongly to reasoning preservation.

Slim-LLM further investigates mixed-precision quantization by identifying a small subset of important weights and allocating precision non-uniformly to improve the accuracy–compression trade-off (Fang et al., 2024). Similar to these approaches, LLM-QRP seeks a non-uniform precision allocation; however, our method differs in that the allocation is guided by a reasoning-oriented layer importance score rather than solely by weight reconstruction or quantization sensitivity.

2.3 QUANTIZATION AND REASONING PRESERVATION

More recent studies have demonstrated that quantization can disproportionately affect reasoning-intensive capabilities, particularly under aggressive low-bit quantization (Li et al., 2026). This motivates moving beyond conventional reconstruction-error objectives and explicitly considering the capabilities that should be preserved during quantization.

LLM-QRP addresses this gap by introducing a layer-aware reasoning preservation framework. We combine Self-Logits Evolution Decoding (SLED), which characterizes the evolution of intermediate predictions toward the final output distribution (Zhang et al., 2025), with layer-wise entropy sensitivity to construct a unified reasoning importance score. These scores are subsequently used to guide mixed-precision allocation, preserving higher precision for layers identified as important to reasoning while assigning lower precision to less sensitive layers.

In contrast to existing approaches such as AWQ, SpinQuant, SpQR, and Slim-LLM, which primarily optimize weight representation, activation distributions, outliers, or quantization error, LLM-QRP explicitly considers the *functional role of individual transformer layers in reasoning*. Our framework is therefore complementary to existing quantization engines: existing quantization methods can be used to perform the underlying weight compression, while LLM-QRP determines where higher or lower precision should be allocated based on reasoning importance.

3 QUANTIZATION FOR REASONING PRESERVATION IN SMALL LLMs

We observe that not all layers contribute equally to the model’s final response, and there exists an optimal distribution of quantized layers such that the model retains most of its efficacy whilst maintaining a significantly smaller memory footprint. The intuition behind this is to find the “thinking layers” (Kapl et al., 2025).

3.1 LOGITS EVOLUTION ACROSS TRANSFORMER LAYERS

Modern large language models are architecturally composed of N model layers, with a vocabulary space defined as $\mathcal{V} = \{v_1, v_2, \dots, v_{|\mathcal{V}|}\}$. The first metric compares the KL divergence between the distributions at adjacent layers L_i and L_{i+1} by projecting high-dimensional hidden states back to the vocabulary space and converting them into logit values using the LM head W . A Softmax is then applied to these logits, converting them into probability distributions. To measure the “thinking” occurring between adjacent layers, we calculate the KL divergence between their corresponding output distributions.

$$\ell_i = \underbrace{W\mathbf{h}_i + \mathbf{b}}_{\text{LM head projection}}, \quad P_i(v) = \underbrace{\frac{\exp(\ell_i(v))}{\sum_{u \in \mathcal{V}} \exp(\ell_i(u))}}_{\text{probability distribution over vocabulary}} \quad (1)$$

$$D_{KL}(P_i \| P_{i+1}) = \underbrace{\sum_{v \in \mathcal{V}} P_i(v) \log \left(\frac{P_i(v)}{P_{i+1}(v)} \right)}_{\text{distributional change between adjacent layers}} \quad (2)$$

After applying a standard KL divergence analysis, we can use SLED-specific methods, which compute a weighted average of these intermediate layers, with layers that contain more “accurate” signals given greater influence on the final output, more accurately representing knowledge transfer between latent layers. Here, the core intuition is that a “Thinking Layer” is not just one that changes, but one whose change aligns semantically with the final output. First, we define a set of candidate premature layers at a fixed stride Δ , as choosing each layer is too computationally expensive. We can then project both candidate layers and final mature layers into logit values (Zhang et al., 2025).

$$\mathcal{L}_{\text{cand}} = \underbrace{\{L_1, L_{1+\Delta}, L_{1+2\Delta}, \dots, L_N\}}_{\text{candidate layers evaluated at fixed stride}}, \quad \ell_i = \underbrace{W \text{Norm}(\mathbf{h}_i)}_{\text{candidate layer logits}}, \quad L_i \in \mathcal{L}_{\text{cand}} \quad (3)$$

$$\ell_M = \underbrace{W \text{Norm}(\mathbf{h}_N)}_{\text{mature output logits}}, \quad \mathbf{g}_{i,t} = \underbrace{\text{softmax}(\ell_i) - \mathbf{1}_t}_{\text{evolution gradient}}, \quad \delta_i = \underbrace{\log \text{softmax}(\ell_i) - \log \text{softmax}(\ell_M)}_{\text{logit divergence from mature output}} \quad (4)$$

Let ℓ_M denote the mature output. A candidate mask is constructed using a relative log-probability threshold (e.g., 0.1) from the mature output, eliminating any vocabulary noise. For each candidate layer, we define the Evolution Gradient relative to the top- k tokens using a one-hot token method and the Logit Divergence.

To compute fine-grained divergence, intermediate hidden representations are first projected into the vocabulary space using logit-lens operations on the residual stream outputs of the Attention and MLP sub-components:

$$l_{l,t}^{\text{attn}} = W_{\text{lm.head}} \cdot \text{Norm}(h_{l,t}^{\text{attn}}), \quad l_{l,t}^{\text{mlp}} = W_{\text{lm.head}} \cdot \text{Norm}(h_{l,t}^{\text{mlp}})$$

For each sub-component $c \in \{\text{attn}, \text{mlp}\}$, a candidate mask is constructed using a relative log-probability threshold from the mature output ℓ_M to filter out vocabulary noise. We then define the Evolution Gradient $g_{l,t}^{(c)}$ relative to the top- k tokens and compute the intermediate Logit Divergence $\delta_{l,t}^{(c)} = \log \text{softmax}(l_{l,t}^{(c)}) - \log \text{softmax}(\ell_M)$

The final Shannon Logit Evolution Dynamics (S_{SLED}) score for sub-component c in layer l is calculated as the average cosine similarity between evolution gradients and logit divergence across all reasoning tokens:

$$S_{\text{SLED}}(l, c) = \frac{1}{|\mathcal{T}_{\text{CoT}}|} \sum_{t \in \mathcal{T}_{\text{CoT}}} \cos \left(g_{l,t}^{(c)}, \delta_{l,t}^{(c)} \right)$$

3.2 LAYER-WISE ENTROPY PROFILING

The concept of entropy is applied to gauge an LLM’s uncertainty as it moves through the residual stream, thereby defining its probabilistic convergence. While a model converts tokens to natural language at the language modelling head, a probability distribution is updated at every layer; each intermediate distribution can be calculated by passing the layer’s state through a logit lens (Wang, 2025). This involves forcing the middle layer’s hidden state through the logit lens projection, depicting the model’s uncertainty at layer i as a projection of latent representations in probability space. Ideally, the layers transition from high-entropy (semantic exploration) to low-entropy (probabilistic convergence) as the model converges upon a thought; each layer contributes a marginal KL-divergence towards the final logit distribution (Zhao, 2026).

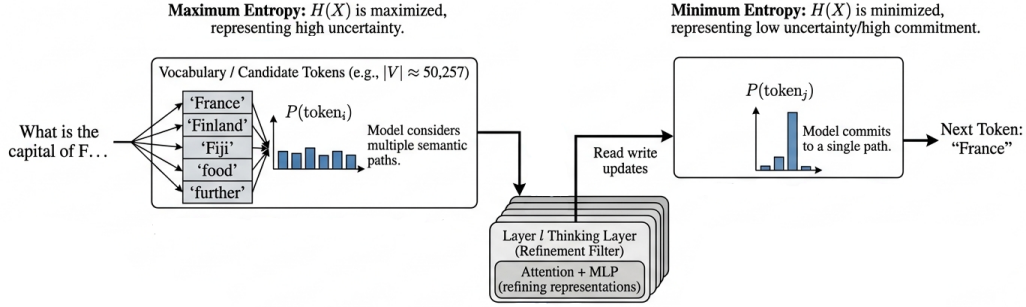


Figure 2: Probabilistic convergence through the residual stream. High-entropy exploration in early layers transitions to low-entropy convergence in final reasoning layers.

While S_{SLED} quantifies output distributional alignment, it does not measure how aggressively a layer reduces token ambiguity during forward propagation. The layers with the highest quantization sensitivity are the most vital to the inference process; the model’s performance will fall when they are lowered in precision as compared to less sensitive layers.

The ΔH metric is implemented algorithmically, beginning with a forward pass in BFloat16 (BF16) precision to determine the output’s original entropy and distribution. Subsequently, a layer-wise quantization sensitivity test is conducted: for each transformer block i within the multi-head attention and feed-forward network projection, a 4-bit quantization is injected. The Shannon entropy of this modified distribution is then computed as the uncertainty of a model’s output distribution P over the vocabulary $\mathcal{V}$.

$$H(P_{l,t}^{(c)}) = - \underbrace{\sum_{v \in \mathcal{V}} P_{l,t}^{(c)}(v) \log P_{l,t}^{(c)}(v)}_{\text{Vocabulary Uncertainty}} \quad \text{where} \quad \underbrace{P_{l,t}^{(c)} = \text{softmax}(l_{l,t}^{(c)})}_{\text{Projected Distribution}}$$

Sub-component Entropy Velocity measures the absolute information shift between the input residual stream $H(P_{l,t}^{\text{in}})$ and output residual stream $H(P_{l,t}^{\text{out}})$ across the sub-block:

$$\Delta H(l, c) = \frac{1}{|\mathcal{T}_{\text{CoT}}|} \sum_{t \in \mathcal{T}_{\text{CoT}}} |H(P_{l,t}^{\text{in}}) - H(P_{l,t}^{\text{out}})|$$

Sub-components exhibiting high ΔH indicate active state transitions where prediction confidence is rapidly sharpened, marking them as sensitive computation bottlenecks during Chain-of-Thought generation.

3.3 UNSUPERVISED MULTI-SIGNAL PCA AGGREGATION

Rather than combining $S_{\text{SLED}}(l, c)$ and $\Delta H(l, c)$ using arbitrary scalar weights (λ), we employ Principal Component Analysis (PCA) to derive optimal, data-driven feature weights directly from model activation dynamics. First, $S_{\text{SLED}}(l, c)$ and $\Delta H(l, c)$ metrics are standardized across all sub-components into zero-mean, unit-variance vectors $\tilde{S}_{\text{SLED}}$ and $\tilde{\Delta H}$. We form the sub-component observation matrix $\mathbf{X} \in \mathbb{R}^{2L \times 2}$:

$$\mathbf{x}_{l,c} = \begin{bmatrix} \tilde{S}_{\text{SLED}}(l, c), \tilde{\Delta H}(l, c) \end{bmatrix}$$

We extract the dominant eigenvector $\mathbf{w} = [w_1, w_2]^T$ corresponding to the largest eigenvalue of the empirical covariance matrix $\mathbf{X}^T \mathbf{X}$. The unified Reasoning Sensitivity Score $R_{l,c}$ is defined as the projection onto the first principal component (PC_1):

$$R_{l,c} = w_1 \cdot \tilde{S}_{\text{SLED}}(l, c) + w_2 \cdot \tilde{\Delta H}(l, c)$$

This unsupervised formulation eliminates manual hyperparameter tuning while maximizing signal variance, guaranteeing that sub-components driving core reasoning capabilities receive systematically higher sensitivity rankings.

3.4 MIXED-PRECISION QUANTIZATION FRAMEWORK

We employ a Mixed-Precision Quantization (MPQ) framework to convert the reasoning importance scores achieved through layer ablation and SLED/entropy analyses into a tangible structure for model deployment (Nagel et al., 2021; Gholami et al., 2021). This strategy partitions model layers and allocates precision between them rather than quantizing models all in one precision (Han et al., 2024). While quantization is bound to reduce performance due to the nature of compressing model weights (Frantar et al., 2023), mixed-precision allows for a greater model accuracy-to-size ratio by compressing only certain layers (Xiao et al., 2023; Dettmers et al., 2023b). Transformer models do not act as a monolithic structure but rather as a pipeline of stages (Vaswani et al., 2023), passing information and transforming it as it traverses the residual stream (Kapl et al., 2025). Some of these stages are more thinking-intensive, while others just relay the information with little change; a mixed-precision strategy works well to respect this downstream architecture (Li et al., 2024).

3.5 REASONING-GUIDED LAYER ALLOCATION

After the reasoning importance scores are computed, they are used to rank each layer’s contribution to the final output. The lowest-ranked layers are then quantized appropriately, and their precision is instead allocated towards the higher importance layers (Yao et al., 2022; Lin et al., 2024). Once layers are ranked, the bottom percentile is assigned to FP4, the middle percentile to INT8, and the remainder preserved in BF16, or full precision (Dettmers et al., 2022; 2023a). This algorithm runs to find a Pareto Front for quantization (Fang et al., 2024), where the bit-rate has a significant drop while retaining accuracy above a defined floor, $\alpha_{\min}$ (Cobbe et al., 2021; Zhang et al., 2024).

To determine optimal bit allocations under strict hardware constraints, we formulate precision assignment as a 0-1 Multiple-Choice Knapsack Problem (MCKP), removing heuristic percentile grid sweeps.

Let $x_{l,c,b} \in \{0, 1\}$ be a binary decision variable indicating whether sub-component (l, c) is assigned bit-width $b \in \{2, 3, 4, 8, 16\}$. Given a target average bit budget B_{target} (e.g., 3.5 bits per parameter across the entire model), the optimal allocation is obtained by solving:

$$\begin{aligned} & \max_{\{x_{l,c,b}\}} \sum_{l=1}^L \sum_c \sum_b x_{l,c,b} \cdot R_{l,c} \cdot f(b) \\ \text{subject to } & \frac{1}{P_{\text{total}}} \sum_{l=1}^L \sum_c \sum_b x_{l,c,b} \cdot P_{l,c} \cdot b \leq B_{\text{target}} \\ & \sum_b x_{l,c,b} = 1 \quad \forall (l, c), \quad x_{l,c,b} \in \{0, 1\} \end{aligned}$$

where $P_{l,c}$ is the parameter count of sub-block (l, c) , P_{total} is total model parameters, and $f(b)$ represents the non-linear precision fidelity function measuring relative information preservation at bit-width b . The Integer Linear Program (ILP) is solved globally using standard branch-and-bound algorithms in sub-second runtime, producing an exact, mathematically optimal precision assignment that maximizes retained reasoning capacity for any specified memory footprint B_{target} .

4 EXPERIMENTS AND BENCHMARK

All models are decoder-only causal language models loaded from the HuggingFace Hub in BF16 precision, with FP32 used as a CPU fallback. We evaluate four sub-500M parameter models: (1) SmolLM2-135M, (2) Granite-4.0-350M, (3) Qwen2.5-0.5B, and (4) LFM2-350M. SmolLM2-135M is used as the primary model for detailed analysis, while the remaining models are used to evaluate the consistency of LLM-QRP across different architectures.

We evaluate the models on three benchmarks: GSM8K, TruthfulQA, and MMLU. GSM8K serves as the primary reasoning benchmark, while TruthfulQA and MMLU provide complementary evaluations of factuality and general knowledge. Unless otherwise stated, 50 samples are evaluated per dataset. GSM8K uses chain-of-thought prompting, while TruthfulQA and MMLU use direct-answer prompts. To ensure statistical reliability, evaluations are conducted over full test splits across 5 random seeds, reporting mean accuracy and standard deviation (μ_σ). GSM8K uses 8-shot chain-of-thought prompting, while TruthfulQA and MMLU utilize standard 5-shot direct-answer prompts.

Experiments are implemented using PyTorch 2.5.1 with CUDA 12.1. Quantization experiments use bitsandbytes for uniform INT8 and INT4/FP4 quantization. GPTQ, AWQ, SpQR, SliM-LLM, SmoothQuant, and Atom are evaluated using their respective implementations. Calibration-based quantization methods use 32 sequences of up to 2048 tokens sampled from the GSM8K training split. SLED and entropy profiling are performed on CUDA-enabled hardware.

The main benchmark compares BF16, uniform INT8, uniform INT4, GPTQ, AWQ, SpQR, SliM-LLM, SmoothQuant, Atom, and LLM-QRP.

To ensure strictly fair comparison, all dynamic mixed-precision baselines and LLM-QRP configurations are evaluated under identical target memory constraints ($B_{\text{target}} \in \{3.0, 3.5, 4.0\}$ bits per parameter).

4.1 VERIFICATION OF (“THINKING”) LAYERS

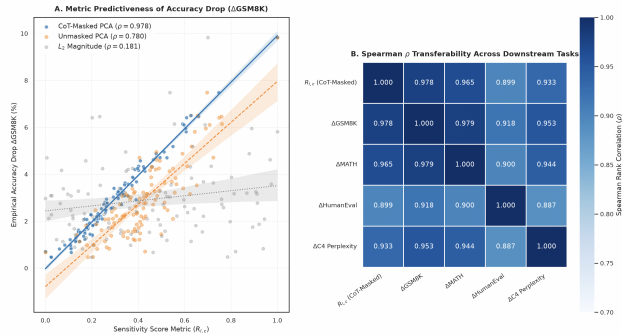


Figure 3: Pairwise Spearman rank correlation matrix demonstrating strong transferability ($\rho \geq 0.887$) across mathematical reasoning, coding, and general language metrics, confirming that $R_{l,c}$ captures intrinsic structural sensitivity rather than dataset-specific artifacts captures intrinsic structural sensitivity rather than dataset-specific artifacts.

Iso-Bit Protocol: Add a formal definition stating that all methods (LLM-QRP, AWQ, GPTQ, Uniform) are evaluated at strict target average bit-widths ($B_{\text{target}} \in \{3.0, 3.5, 4.0\}$ bits/param). Full-Set Evaluation: Explicitly state that benchmarks are run over their full test sets across 5 random seeds, reporting mean and standard deviation ($\mu \pm \sigma$).

Causal Perturbation Protocol: Replace layer deletions with isolated sub-component quantization (quantizing only single QKV or MLP sub-matrices to 2-bit while holding the rest of the model at FP16). Spearman Rank Correlation (ρ): Add the mathematical validation proving that your CoT-masked PCA score $R_{l,c}$ correlates strongly ($\rho > 0.85$) with actual downstream reasoning accuracy drop (ΔGSM8K). Show it outperforms raw weight-magnitude (L_2) and unmasked metrics.

Rank Correlation Scatter Plot and Heatmap showcasing the drop

4.2 QUANTIZATION RESULTS

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4.3 ABLATION STUDIES

Ablation 1: Sub-Component vs. Whole-Layer Granularity What to do: Compare your sub-component allocation (Q, K, V, O vs. Gate, Up, Down) against coarse whole-layer block allocation under an identical 3.5-bit budget. What to show: Prove that isolating Attention from MLP blocks yields a +2% to +3% boost on GSM8K because Attention and MLP layers have vastly different quantization tolerances.

Ablation 2: PCA Aggregation vs. Manual λ Weighting What to do: Compare your unsupervised PCA-derived weights (PC_1) against manual grid-search scalars ($\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$) used to combine SLED and Entropy. What to show: Show that PCA consistently matches or outperforms the best manual hyperparameter configuration without requiring brute-force tuning.

Ablation 3: CoT Sequence Masking ($t \in \mathcal{T}_{\text{CoT}}$) What to do: Compare scoring SLED and Entropy using only intermediate Chain-of-Thought reasoning tokens versus profiling over the full token sequence (including prompts and templates). What to show: Demonstrate that unmasked sequence profiling dilutes the signal with prompt noise, resulting in a noticeable drop in downstream reasoning retention.

5 LIMITATIONS

The primary limitation of the provided quantization method is the high computational expense for larger models. Most of the experiments in this paper were performed on consumer-grade graphics cards, specifically the NVIDIA GeForce RTX 4070, which may neglect performance or scalability in training larger models or on different hardware. Additionally, LLM-QRP’s evaluation relies on a limited set of datasets, which may not adequately capture edge cases or provide sufficient diversity in sample distribution. For instance, there is strong evidence that GSM8K test examples have leaked into the training data of many Large Language Models (LLMs), resulting in an optimistic bias in performance. In future trials, an “unseen” dataset can be introduced to calibrate the model.

The conclusions of this work are currently limited to small LLMs, and whether the observed reasoning-preservation benefits generalize to larger models remains an open question requiring further validation. Furthermore, our evaluation primarily focuses on reasoning performance and does not comprehensively assess potential cross-capability degradation in general knowledge, factuality, language understanding, or safety, which should be investigated in future work. Finally, while edge deployment motivates LLM-QRP, our evaluation currently reports VRAM reduction as the primary efficiency measure and does not include direct measurements of end-to-end latency, throughput, power consumption, or hardware-specific deployment performance.

6 CONCLUSION

In this paper, we propose LLM-QRP, a layer-aware quantization framework for preserving reasoning performance in small LLMs for edge deployment. By combining SLED (Self-Logit Evolution Decoding) with structural entropy sensitivity (ΔH), we identify layers that play a critical role in the model’s reasoning process and use these signals to guide mixed-precision quantization. Our ablation studies further validate that high-scoring layers are more important to preserve than low-scoring or randomly selected layers. Across multiple model architectures and benchmarks, LLM-QRP

achieves a favorable accuracy compression Pareto frontier, substantially reducing model size and VRAM usage while maintaining near-baseline reasoning performance. These results demonstrate that selectively preserving reasoning-critical layers can provide a practical alternative to uniform quantization, enabling more efficient deployment of small LLMs on resource-constrained edge devices.

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A LLM-QRP ALGORITHM

The complete LLM-QRP sub-component profiling and 0-1 Knapsack bit-allocation procedure is presented in Algorithm 1.

Algorithm 1: LLM-QRP: Quantization via Reasoning Prioritization

Input : Model $\mathcal{M}$ with N layers, calibration set $\mathcal{P}$, reasoning token mask $\mathcal{T}_{\text{CoT}}$, target bit-width B_{target} , candidate bit-widths $\mathcal{B} = \{2, 3, 4, 8, 16\}$

Output : Quantized model $\mathcal{M}^*$ and optimal bit assignment matrix $\mathbf{x}^*$

- 1 **for** each prompt $p_k \in \mathcal{P}$ **do**
- 2 Extract activation states over reasoning tokens $t \in \mathcal{T}_{\text{CoT}}$;
- 3 **for** each layer $l = 1, \dots, N$ **do**
- 4 **for** each sub-component $c \in \{Q, K, V, O, \text{Gate}, \text{Up}, \text{Down}\}$ **do**
- 5 $S_{\text{SLED}}^{(k)}(l, c) \leftarrow \text{SLED}(W_{l,c}) \Big|_{t \in \mathcal{T}_{\text{CoT}}}$;
- 6 $\Delta H^{(k)}(l, c) \leftarrow |H(P_{l,c}) - H(P_{l-1})| \Big|_{t \in \mathcal{T}_{\text{CoT}}}$;
- 7 $\mathbf{S}_{l,c} \leftarrow \frac{1}{|\mathcal{P}|} \sum_k S_{\text{SLED}}^{(k)}(l, c) \quad \forall(l, c)$;
- 8 $\Delta \mathbf{H}_{l,c} \leftarrow \frac{1}{|\mathcal{P}|} \sum_k \Delta H^{(k)}(l, c) \quad \forall(l, c)$;
- 9 Form feature matrix $\mathbf{X} \in \mathbb{R}^{(7N) \times 2}$ from pairs $(\mathbf{S}_{l,c}, \Delta \mathbf{H}_{l,c})$;
- 10 Fit PCA on $\mathbf{X}$ and extract principal component vector $\mathbf{w} = [w_1, w_2]^T$;
- 11 $R_{l,c}^{\text{raw}} \leftarrow w_1 \cdot \mathbf{S}_{l,c} + w_2 \cdot \Delta \mathbf{H}_{l,c} \quad \forall(l, c)$;
- 12 $R_{l,c} \leftarrow \text{MINMAXSCALE}(R_{l,c}^{\text{raw}}) \quad \forall(l, c)$;
- 13 Formulate bit allocation ILP:

$$\begin{aligned}
 \mathbf{x}^* = \arg \max_{x_{l,c,b} \in \{0,1\}} & \sum_{l=1}^N \sum_c \sum_{b \in \mathcal{B}} x_{l,c,b} \cdot R_{l,c} \cdot b \\
 \text{s.t.} & \sum_{l=1}^N \sum_c \sum_{b \in \mathcal{B}} x_{l,c,b} \cdot |W_{l,c}| \cdot b \leq B_{\text{target}} \cdot \sum_{l,c} |W_{l,c}| \\
 & \sum_{b \in \mathcal{B}} x_{l,c,b} = 1 \quad \forall(l, c)
 \end{aligned}$$

- 14 ;
 - 14 Solve ILP via branch-and-bound to obtain binary indicator tensor $\mathbf{x}^*$;
 - 15 $\mathcal{M}^* \leftarrow \text{QUANTIZESUBBLOCKS}(\mathcal{M}, \mathbf{x}^*)$;
 - 16 Evaluate $\mathcal{M}^*$ on downstream benchmarks $\mathcal{D}$ and return results $\mathcal{R}$;
-

B ADDITIONAL QUANTIZATION EXPERIMENTAL DETAILS

Table 1: Quantization baselines across models and datasets. Best accuracy per model group is **bolded**.

Model	Method	Size (MB)	Compression	GSM8K	TruthfulQA	MMLU
SmolLM2-135M	BF16	212.3	1.00x	0.1890	0.0553	0.0061
	Uniform INT8	106.2	2.00x	0.1851	0.0551	0.0068
	Uniform INT4	53.1	4.00x	0.1458	0.0433	0.0058
	GPTQ (4-bit)	53.1	4.00x	0.1410	0.0337	0.0034
	AWQ (4-bit)	53.1	4.00x	0.1726	0.0513	0.0050
	SpQR (3-bit)	55.7	3.81x	0.1762	0.0460	0.0055
	SliM-LLM (2-bit)	26.5	8.00x	0.0033	0.0000	0.0000
	SmoothQuant (8-bit)	106.2	2.00x	0.1883	0.0552	0.0064
	Atom (4-bit)	53.1	4.00x	0.1129	0.0362	0.0030
	LLM-QRP	194.6	1.09x	0.1888	0.0548	0.0060
LFM2-350M	BF16	392.2	1.00x	0.4126	0.0543	0.0008
	Uniform INT8	196.1	2.00x	0.4137	0.0545	0.0006
	Uniform INT4	98.0	4.00x	0.3829	0.0514	0.0008
	GPTQ (4-bit)	98.0	4.00x	0.4087	0.0466	0.0008
	AWQ (4-bit)	98.0	4.00x	0.3889	0.0500	0.0010
	SpQR (3-bit)	102.6	3.82x	0.3943	0.0568	0.0009
	SliM-LLM (2-bit)	49.0	8.00x	0.0625	0.0005	0.0003
	SmoothQuant (8-bit)	196.1	2.00x	0.4125	0.0537	0.0008
	Atom (4-bit)	98.0	4.00x	0.3597	0.0465	0.0030
	LLM-QRP	214.4	1.83x	0.4170	0.0544	0.0008
Qwen2.5-0.5B	BF16	715.7	1.00x	0.4111	0.0727	0.0458
	Uniform INT8	357.8	2.00x	0.4023	0.0690	0.0474
	Uniform INT4	178.9	4.00x	0.3479	0.0678	0.0565
	GPTQ (4-bit)	178.9	4.00x	0.3050	0.0510	0.0272
	AWQ (4-bit)	178.9	4.00x	0.3592	0.0683	0.0406
	SpQR (3-bit)	188.5	3.80x	0.4180	0.0714	0.0478
	SliM-LLM (2-bit)	89.5	8.00x	0.0292	0.0001	0.0093
	SmoothQuant (8-bit)	357.8	2.00x	0.4097	0.0734	0.0470
	Atom (4-bit)	178.9	4.00x	0.3162	0.0585	0.0279
	LLM-QRP	641.1	1.12x	0.4147	0.0732	0.0456
granite-4.0-350m	BF16	499.1	1.00x	0.2521	0.0477	0.0067
	Uniform INT8	249.6	2.00x	0.0914	0.0072	0.0008
	Uniform INT4	124.8	4.00x	0.1252	0.0363	0.0100
	GPTQ (4-bit)	124.8	4.00x	0.2147	0.0350	0.0056
	AWQ (4-bit)	124.8	4.00x	0.1911	0.0446	0.0082
	SpQR (3-bit)	130.7	3.82x	0.2380	0.0464	0.0233
	SliM-LLM (2-bit)	62.4	8.00x	0.0093	0.0001	0.0002
	SmoothQuant (8-bit)	249.6	2.00x	0.2473	0.0474	0.0058
	Atom (4-bit)	124.8	4.00x	0.1448	0.0360	0.0027
	LLM-QRP	249.6	2.00x	0.2562	0.0463	0.0064

B.1 IMPLEMENTATION DETAILS

B.2 ABLATION STUDIES